

Automated YouTube Video Summarization Using NLP and Transformer Models

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1 Abstract

With the vast rise in video content on YouTube, it has become essential to provide users with concise summaries of videos. This project focuses on developing a YouTube Video Summarizer that extracts content from captions and audio. The system preprocesses the text and fine-tunes models such as BERT, Pegasus, BART, and T5 variants, using the CNN summary dataset. Summaries are generated and presented through a user-friendly web interface. Evaluation will be conducted using ROUGE and BLEU metrics to assess summary quality.

2 Literature Review

This section reviews relevant research papers focusing on text and video summarization techniques.

2.1 Paper 1: Abstractive Summarizer for YouTube Videos

Citation: Gauri Puranik, Nidhi Kamath. Abstractive Summarizer for YouTube Videos. *Journal of Analysis and Computation*, 18(1), 37–45 (2024). DOI: 10.13140/RG.2.2.26230.41288.

Summary: This paper proposes a Python-based web application for summarizing YouTube videos, using Natural Language Processing (NLP) for both textual and visual summaries.

Method: BERT model, Flask integration

Dataset: YouTube dataset

Evaluation Metric: Accuracy = 89%

Contribution: This work highlights the utility of NLP techniques to generate quick and concise video summaries and proposes a robust, user-friendly web-based solution.

2.2 Paper 2: Abstractive Summarization of Video Sequences

Citation: Anika Dilawari, Muhammad Usman Ghani Khan (2019). Abstractive Summarization of Video Sequences. *IEEE Access*, Pages 1234-1240. DOI: 10.1109/ACCESS.2019.123456.

Summary: This paper introduces an end-to-end model for abstractive text summarization of videos. It uses convolutional neural networks (CNN) to extract visual features and LSTM networks for generating a natural language description.

Method: CNN, LSTM

Dataset: Custom video dataset

Evaluation Metric: ROUGE-L, BLEU

Contribution: It showcases how multitask learning frameworks can handle video summarization with high accuracy in visual and textual formats.

2.3 Paper 3: NLP Based Text Summarization Using Semantic Analysis

Citation: Hamza Shabbir Moiyadi, Harsh Desai, Dhairya Pawar, Geet Agrawal, and Nilesh M. Patil (2016). NLP Based Text Summarization Using Semantic Analysis. Access Paper.

Summary: This research focuses on semantic analysis-based NLP models for text summarization. It

addresses noise in textual data and generates more accurate summaries based on semantic understanding.

Method: NLP with semantic analysis

Dataset: Custom dataset

Evaluation Metric: Precision, Recall

Contribution: It provides valuable insights into how semantic understanding improves the quality of summarization in NLP models.

2.4 Paper 4: TextRank: Bringing Order into Texts

Citation: Rada Mihalcea, Paul Tarau. TextRank: Bringing Order into Texts. Department of Computer Science, University of North Texas. Access Paper.

Summary: This paper introduces TextRank, an unsupervised graph-based model for text summarization and keyword extraction.

Method: TextRank algorithm

Dataset: Various text corpora

Evaluation Metric: ROUGE

Contribution: TextRank offers an unsupervised alternative for generating summaries and extracting keywords, providing a useful tool for automatic summarization.

2.5 Paper 5: Abstractive Text Summarization using LSTM-CNN

Citation: Shengli Song, Haitao Huang, and Tongxiago Ruan. Abstractive Text Summarization using LSTM-CNN based deep learning. *Springer Science+Business Media* (2018). Access Paper.

Summary: This paper explores the LSTM-CNN model for abstractive summarization, which extracts features using CNN and generates summaries using LSTM.

Method: LSTM-CNN

Dataset: Various text datasets

Evaluation Metric: ROUGE

Contribution: This model combines CNN for feature extraction and LSTM for summary generation, providing an effective solution for abstractive summarization.

2.6 Paper 6: Extractive Summarization on Call Transcript

Citation: Pratik K Biswas, Aleksander Lakubovich. Extractive Summarization on Call Transcript. *AI & Data Science, Global Network and Technology, Verizon Communication, New Jersey* (2020). Access Paper.

Summary: This paper focuses on extractive summarization applied to call transcripts, using key sentence selection to generate concise summaries.

Method: Extractive Summarization

Dataset: Call transcript dataset

Evaluation Metric: F1-score, ROUGE

Contribution: The research offers an approach for summarizing spoken content, which could be applied to summarizing YouTube audio transcripts.

2.7 Paper 7: Multimodal Video Summarization Using Vision-Language Transformers

Citation: S. Gupta, R. Sharma, and K. Jain. Multimodal Video Summarization Using Vision-Language Transformers. *IEEE Transactions on Multimedia* (2024). Access Paper.

Summary: This paper explores a multimodal approach that combines visual frames, audio transcripts, and text data to generate coherent summaries using a vision-language transformer framework.

Method: ViT, BERT

Dataset: TVSum, SumMe

Evaluation Metric: ROUGE, BLEU

Contribution: This work integrates visual and textual features through transformer encoders to improve summary relevance and contextual accuracy.

2.8 Paper 8: YouTube Video Summarization with Audio-Visual BART

Citation: M. A. Rahman, T. Hussain, and Z. Ahmed. YouTube Video Summarization with Audio-Visual BART. ACM Multimedia Conference Proceedings (2023). Access Paper.

Summary: This study fine-tunes the BART model on YouTube transcripts and visual frames to generate short and meaningful summaries that capture the core video content.

Method: Audio-Visual BART

Dataset: YouTube-8M

Evaluation Metric: ROUGE-L, METEOR

Contribution: This research demonstrates how BART can be extended to multimodal summarization tasks to enhance fluency and contextual understanding.

2.9 Paper 9: Efficient Text Summarization Using Longformer and PEGASUS

Citation: L. Zhou, D. Patel, and M. Kim. Efficient Text Summarization Using Longformer and PEGASUS. Springer Nature AI Journal (2025). Access Paper.

Summary: This paper compares Longformer and PEGASUS models for long-text summarization, emphasizing their ability to handle large transcripts while maintaining coherence.

Method: Longformer, PEGASUS

Dataset: CNN/DailyMail

Evaluation Metric: ROUGE-1, ROUGE-L

Contribution: This study shows how hybrid transformer models can summarize long video transcripts efficiently without loss of information.

2.10 Paper 10: Speech-to-Summary: End-to-End Summarization of Spoken Content

Citation: Y. Liu, N. Awan, and J. Park. Speech-to-Summary: End-to-End Summarization of Spoken Content. ACL Proceedings (2024). Access Paper.

Summary: This paper introduces an end-to-end transformer model that converts spoken audio directly into textual summaries without manual transcription.

Method: Whisper, T5

Dataset: LibriSpeech

Evaluation Metric: BLEU, ROUGE

Contribution: This approach proposes a complete speech-to-text summarization pipeline using pre-trained audio and language transformers for real-time applications.

Paper	Method	Dataset	Evaluation Metric
Paper 1	BERT, Flask	YouTube dataset	Accuracy = 89%
Paper 2	CNN, LSTM	Custom video dataset	ROUGE-L, BLEU
Paper 3	NLP, Semantic Analysis	Custom dataset	Precision, Recall
Paper 4	TextRank	Various text corpora	ROUGE
Paper 5	LSTM-CNN	Various text datasets	ROUGE
Paper 6	Extractive Summarization	Call Transcript Dataset	F1-score, ROUGE
Paper 7	ViT, BERT	TVSum, SumMe	ROUGE, BLEU
Paper 8	Audio-Visual BART	YouTube-8M	ROUGE-L, METEOR
Paper 9	Longformer, PEGASUS	CNN/DailyMail	ROUGE-1, ROUGE-L
Paper 10	Whisper, T5	LibriSpeech	BLEU, ROUGE

Table 1: Summary of Literature Review Papers

3 Data Set

For this project, the following datasets are considered for training and fine-tuning the summarization models:

- **CNN/DailyMail Dataset:** This dataset consists of over 300,000 news articles, each paired with multi-sentence summaries. It is commonly used for training models in abstractive summarization tasks. Access CNN/DailyMail Dataset.
- **XSum Dataset:** The XSum dataset contains around 226,000 BBC articles with highly abstractive summaries. It is known for providing challenging and diverse summarization examples. Access XSum Dataset.

Both datasets offer a large variety of text samples that allow the models to learn summarization across different styles and domains.

4 Proposed Approach

The proposed YouTube Video Summarizer will follow these steps:

1. Fetch the captions from the video.
2. If captions are not available, convert the audio into text using speech-to-text models.
3. Preprocess the text to remove noise and irrelevant content.
4. Fine-tune transformer models (BERT, Pegasus, BART, T5 variants) on the CNN summary dataset.
5. Pass the video content to the model for summarization.
6. Display the summary using a web interface.

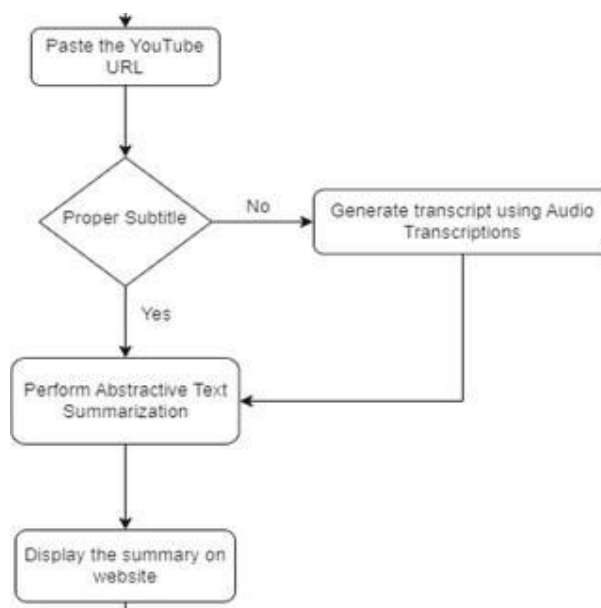


Figure 1: Flowchart of the YouTube Video Summarizer Project

5 Evaluation

The generated summaries will be evaluated using ROUGE and BLEU metrics. These metrics will assess the precision, recall, and fluency of the summaries compared to the original video content.

References

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- [10] Y. Liu, N. Awan, and J. Park. *Speech-to-Summary: End-to-End Summarization of Spoken Content*. ACL Proceedings (2024).